**Practical 9:**

**Secure User Registration with Database Insert, Duplicate Email Check, and Password Hashing**

**Problem Definition Implement secure user registration by inserting user details into MySQL using MYSQLi. Validate input on frontend and backend, check duplicate email/username, hash passwords using password_hash(), and display appropriate messages.**

**CODE:-**

Create a database:

```
CREATE DATABASE user_registration;
```

Then select that database and run:

```
CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50) NOT NULL UNIQUE,
    email VARCHAR(100) NOT NULL UNIQUE,
    password VARCHAR(255) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

**INDEX.PHP**

```
<!DOCTYPE html>
<html>
```

```
<head>
   <title>User Registration</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">

   <h2>User Registration</h2>

   <form action="register.php" method="POST">

      <label>Username</label>
      <input type="text" name="username" required minlength="3" maxlength="50">

      <label>Email</label>
      <input type="email" name="email" required>

      <label>Password</label>
      <input type="password" name="password" required minlength="8">

      <label>Confirm Password</label>
      <input type="password" name="confirm_password" required minlength="8">
```

```
        <button type="submit">Register</button>


    </form>


</div>


</body>
</html>
```

**REGISTRCTION .PHP:-**

```
<?php


include "db.php";


if ($_SERVER["REQUEST_METHOD"] != "POST") {
    header("Location: index.php");
    exit();
}


$username = trim($_POST["username"]);
$email = trim($_POST["email"]);
$password = $_POST["password"];
$confirm_password = $_POST["confirm_password"];


if ($username == "" || $email == "" || $password == "" || $confirm_password
== "") {
```

```
    die("All fields are required");
}

if (strlen($username) < 3 || strlen($username) > 50) {
    die("Username must be between 3 and 50 characters");
}

if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    die("Invalid email address");
}

if (strlen($password) < 8) {
    die("Password must contain at least 8 characters");
}

if ($password != $confirm_password) {
    die("Passwords do not match");
}
$stmt=$conn->prepare("SELECT id from users WHERE username= ? OR email =
?");
$stmt->bind_param("ss", $username, $email);
$stmt->execute();

$result = $stmt->get_result();
```

```
if ($result->num_rows > 0) {
   die("Username or email already exists");
}

$hashed_password = password_hash($password, PASSWORD_DEFAULT);

$stmt = $conn->prepare("INSERT INTO users (username, email, password)
VALUES (?, ?, ?)");
$stmt->bind_param("sss", $username, $email, $hashed_password);

if ($stmt->execute()) {
   echo "Registration successful";
} else {
   echo "Registration failed";
}

$stmt->close();
$conn->close();

?>
```

**db.PHP:-**

```
<?php

$conn = new mysqli("localhost", "root", "", "user_registration");
```

```
if ($conn->connect_error) {
    die("Database connection failed");
}

$conn->set_charset("utf8mb4");

?>
```

**Style.css:-**

```
body {
    font-family: Arial;
    background: #f2f2f2;
}

.container {
    width: 350px;
    margin: 100px auto;
    background: white;
    padding: 25px;
    border-radius: 10px;
}

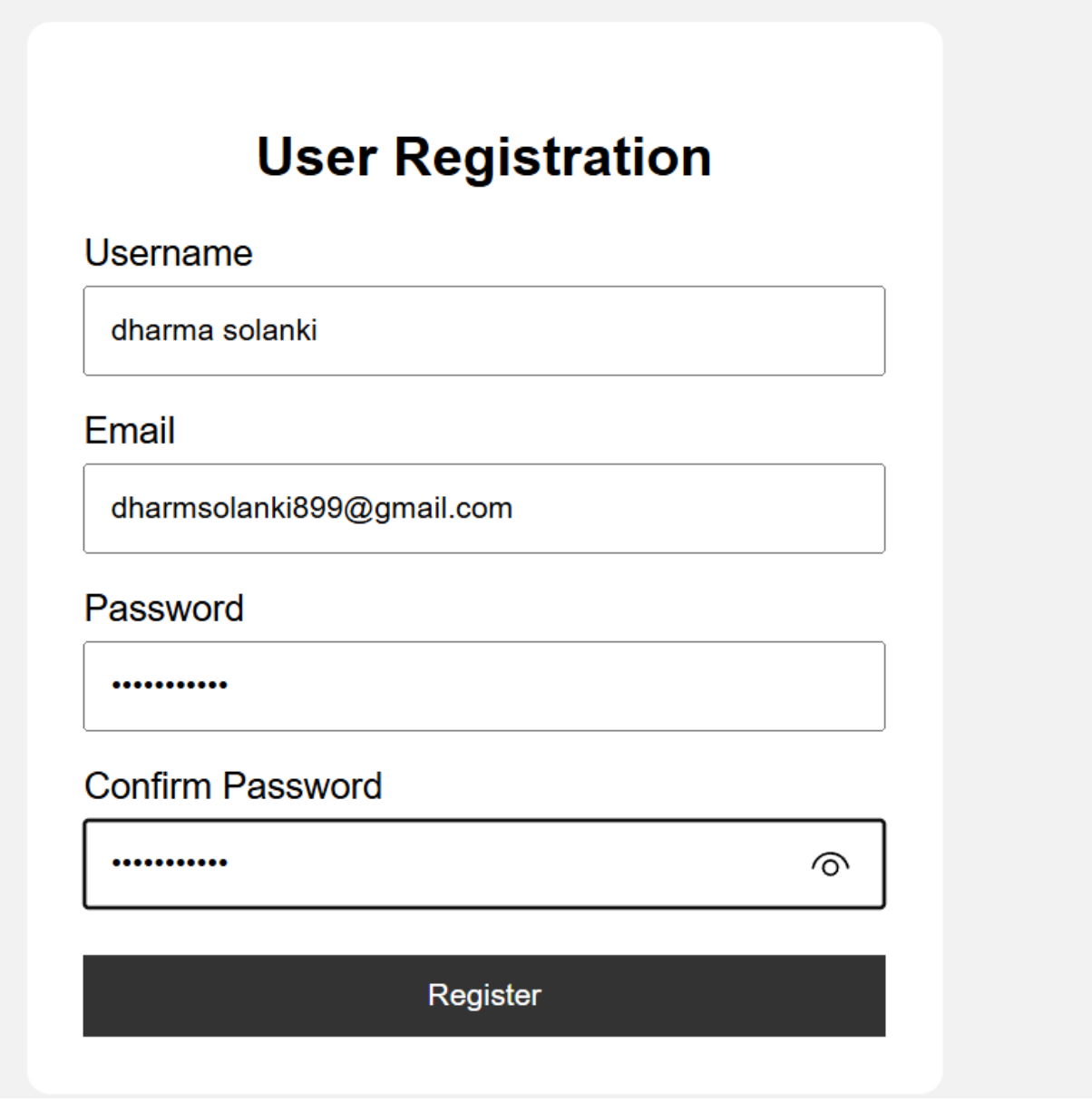
h2 {
    text-align: center;
}
```

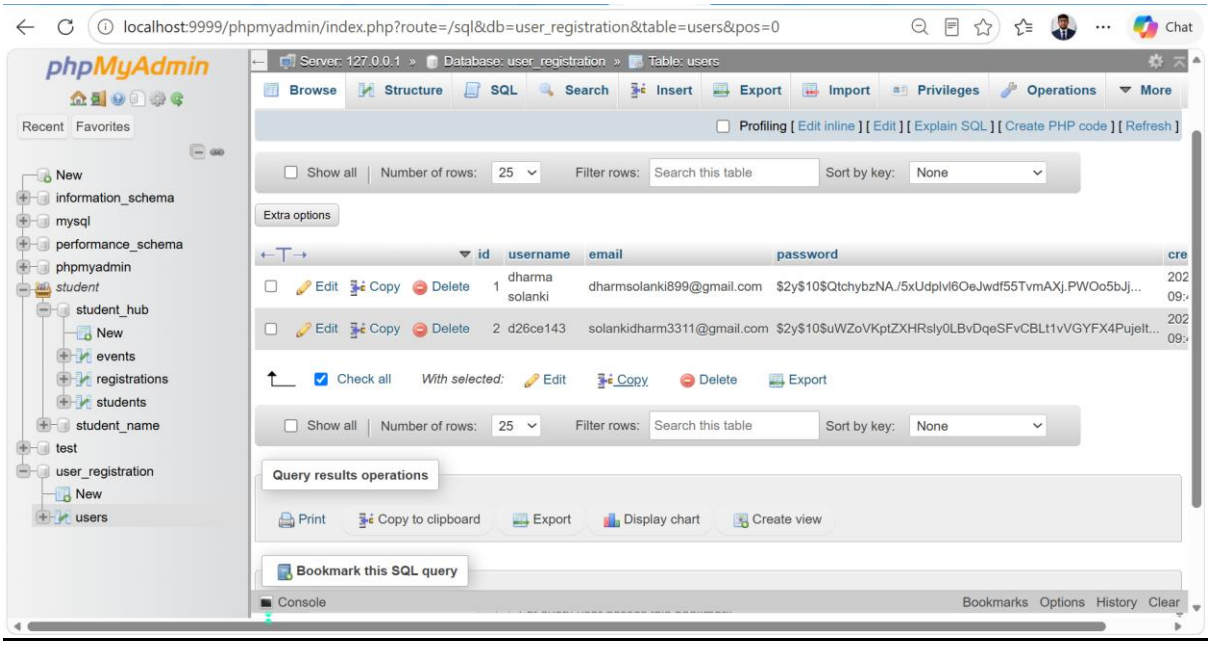
```
label {
    display: block;
    margin-top: 15px;
}

input {
    width: 100%;
    padding: 10px;
    margin-top: 5px;
    box-sizing: border-box;
}

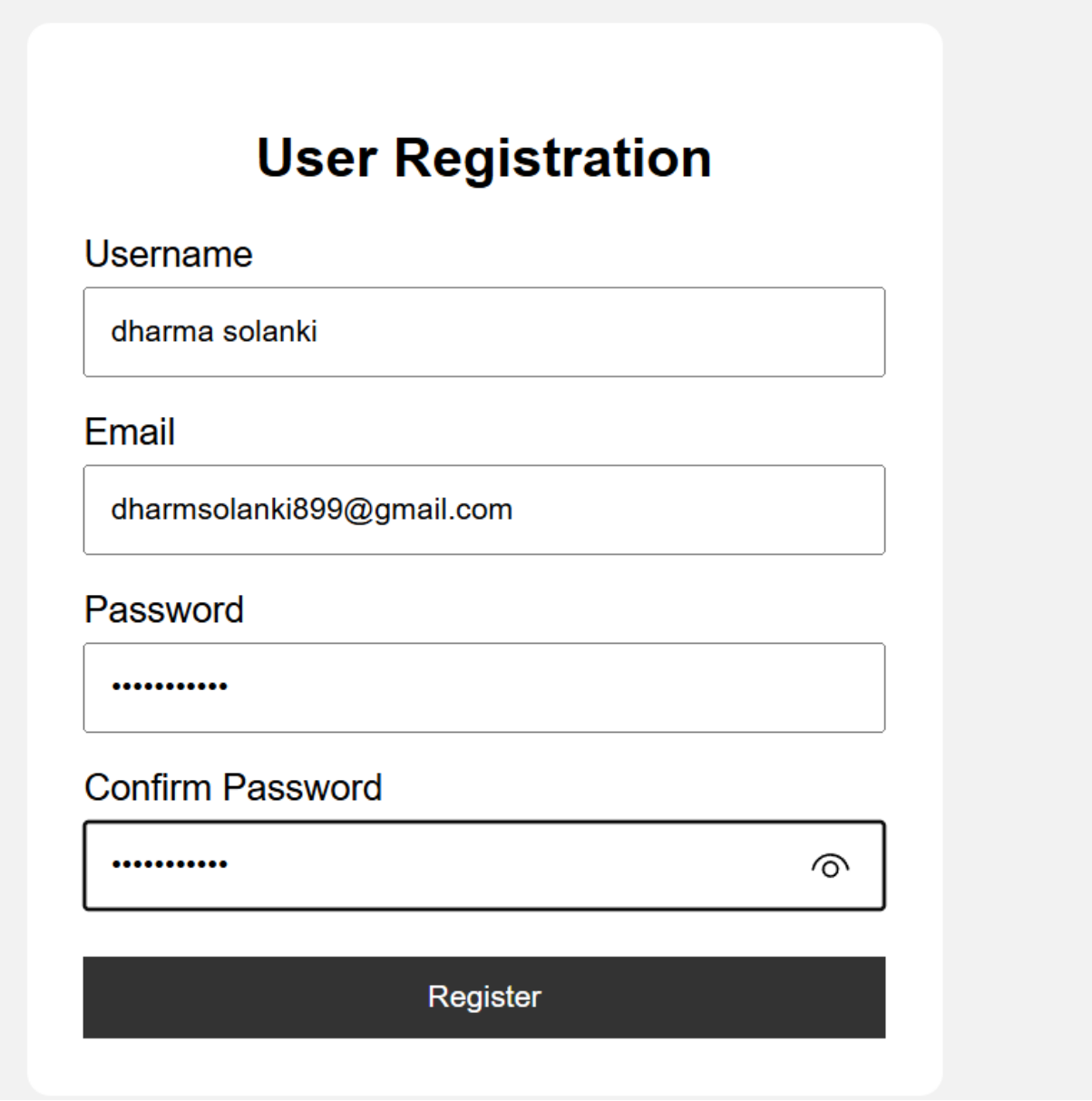
button {
    width: 100%;
    padding: 10px;
    margin-top: 20px;
    background: #333;
    color: white;
    border: none;
    cursor: pointer;
}
```

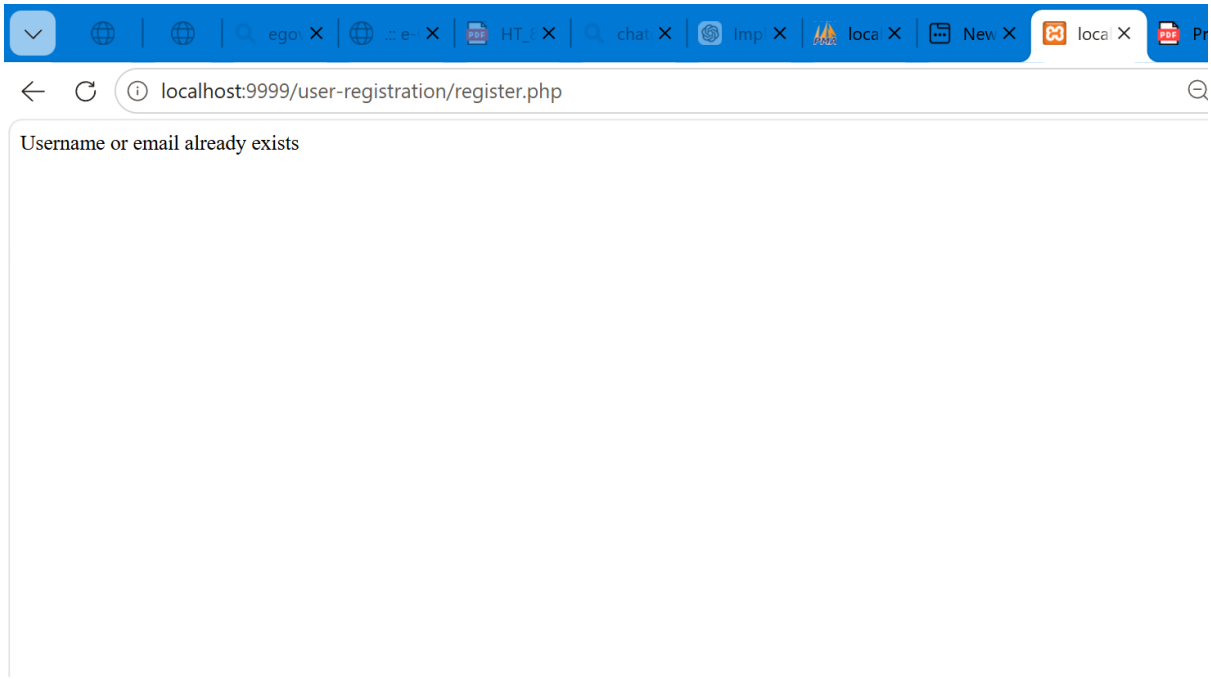
**Output:-**


User Registration
Username
dharma solanki
Email
dharmsolanki899@gmail.com
Password
Confirm Password
Register


localhost:9999/phpmyadmin/index.php?route=/sql&db=user_registration&table=users&pos=0
Chat
phpMyAdmin
Recent Favorites
New
information_schema
mysql
performance_schema
phpmyadmin
student
student_hub
New
events
registrations
students
student_name
test
user_registration
New
users
Server: 127.0.0.1 » Database: user_registration » Table: users
Browse Structure SQL Search Insert Export Import Privileges Operations More
Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
id username email password cre
Edit Copy Delete 1 dharma solanki dharmsolanki899@gmail.com $2y$10$QtchybzNA./5xUdplvl6OeJwdf55TvmAXj.PWOo5bJj... 202
Edit Copy Delete 2 d26ce143 solankidharm3311@gmail.com $2y$10$uWZoVKptZXHRsly0LBvDqeSFvCBLt1vVGYFX4PujeIt... 202
Check all With selected: Edit Copy Delete Export
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Query results operations
Print Copy to clipboard Export Display chart Create view
Bookmark this SQL query
Console
Bookmarks Options History Clear

**If that and that entry**






## Key Questions / Analysis / Interpretation

1. **Is password_hash() used before storing passwords?**
   Yes. password_hash() is used to encrypt/hash the password before storing it in the MySQL database.

2. **Is duplicate email/username checked?**
   Yes. The PHP code checks whether the username or email already exists before inserting a new user.

3. **Are frontend and backend validations consistent?**
   Yes. Frontend validation checks required fields, email format, and password length. Backend validation checks the same requirements again for security.

## Tools/Technology to be Used:-

PHP, MySQLi, XAMPP,VS code